



Công thức kiểm định Xác suất thống kê

Xác suất thống kê (Học viện Công nghệ Bưu chính Viễn thông)



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Kiểm định tham số	Điều kiện	Bài toán	Tiêu chuẩn kiểm định	Miền bác bỏ
Kỳ vọng	Đã biết phương sai	H0: $\mu = \mu_0$ H1: $\mu \neq \mu_0$	$T = \frac{(\bar{X} - \mu_0)\sqrt{n}}{\sigma}$	$W_\alpha = (-\infty, -U_{\alpha/2}) \cup (U_{\alpha/2}, +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu > \mu_0$		$W_\alpha = (U_\alpha, +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu < \mu_0$		$W_\alpha = (-\infty, -U_\alpha)$
	Chưa biết phương sai Kích thước mẫu: $n \geq 30$	H0: $\mu = \mu_0$ H1: $\mu \neq \mu_0$	$T = \frac{(\bar{X} - \mu_0)\sqrt{n}}{S}$	$W_\alpha = (-\infty, -U_{\alpha/2}) \cup (U_{\alpha/2}, +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu > \mu_0$		$W_\alpha = (U_\alpha, +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu < \mu_0$		$W_\alpha = (-\infty, -U_\alpha)$
	Chưa biết phương sai Kích thước mẫu $n < 30$	H0: $\mu = \mu_0$ H1: $\mu \neq \mu_0$	$T = \frac{(\bar{X} - \mu_0)\sqrt{n}}{S}$	$W_\alpha = (-\infty, -t_{\alpha/2}(n-1)) \cup (t_{\alpha/2}(n-1), +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu > \mu_0$		$W_\alpha = (t_\alpha(n-1), +\infty)$
		H0: $\mu = \mu_0$ H1: $\mu < \mu_0$		$W_\alpha = (-\infty, -t_\alpha(n-1))$

Phương sai		$H_0: \sigma^2 = \sigma_0^2$ $H_1: \sigma^2 \neq \sigma_0^2$	$T = \frac{(n-1)S^2}{\sigma^2}$	$W_\alpha = \left(0, \chi_{1-\frac{\alpha}{2}}^2(n-1)\right) \cup \left(\chi_{\frac{\alpha}{2}}^2(n-1), +\infty\right)$
		$H_0: \sigma^2 = \sigma_0^2$ $H_1: \sigma^2 > \sigma_0^2$		$W_\alpha = \left(\chi_\alpha^2(n-1), +\infty\right)$
		$H_0: \sigma^2 = \sigma_0^2$ $H_1: \sigma^2 < \sigma_0^2$		$W_\alpha = \left(0; \chi_{1-\alpha}^2(n-1)\right)$
Tần suất		$H_0: p = p_0$ $H_1: p \neq p_0$	$\begin{cases} np_0 > 5 \\ n(1-p_0) > 5 \end{cases}$ $T = \frac{(f-p)\sqrt{n}}{\sqrt{p(1-p)}}$	$W_\alpha = (-\infty, -U_{\alpha/2}) \cup (U_{\alpha/2}, +\infty)$
		$H_0: p = p_0$ $H_1: p > p_0$		$W_\alpha = (U_\alpha, +\infty)$
		$H_0: p = p_0$ $H_1: p < p_0$		$W_\alpha = (-\infty, -U_\alpha)$
2 Kỳ vọng	Đã biết 2 phương sai σ_1^2 và σ_2^2	$H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$	$T = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$W_\alpha = (-\infty, -U_{\alpha/2}) \cup (U_{\alpha/2}, +\infty)$
		$H_0: \mu_1 = \mu_2$ $H_1: \mu_1 > \mu_2$		$W_\alpha = (U_\alpha, +\infty)$
		$H_0: \mu_1 = \mu_2$ $H_1: \mu_1 < \mu_2$		$W_\alpha = (-\infty, -U_\alpha)$

	Chưa biết 2 phương sai nhưng biết chúng = nhau $1^2=2^2$	H0: $\mu_1 = \mu_2$ H1: $\mu_1 \neq \mu_2$	$T = \frac{(\overline{X}_1 - \overline{X}_2)}{Sp \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $Sp = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$	$W_\alpha = (-\infty, -t_{\alpha/2}(n_1 + n_2 - 2)) \cup (t_{\alpha/2}(n_1 + n_2 - 2), +\infty)$
		H0: $\mu_1 = \mu_2$ H1: $\mu_1 > \mu_2$		$W_\alpha = (t_\alpha(n_1 + n_2 - 2), +\infty)$
		H0: $\mu_1 = \mu_2$ H1: $\mu_1 < \mu_2$		$W_\alpha = (-\infty, -t_\alpha(n_1 + n_2 - 2))$